EvenUCan

from idea to object

User Manual · version 1.9.42

Draw your own things and print them. Or pick the parts for an electronics project, and EvenUCan designs a printable box to hold them, shows you how to wire them up, and writes the code to run them.

It all runs on your own computer. Only the optional AI features use the internet.

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1 Welcome: what can EvenUCan do for me?

EvenUCan takes you from an idea to a finished, working gadget. Use as much or as little of it as you like: each step leads naturally into the next.

1. **Draw your own things.** Sketch a flat shape, pull it up into 3D, round off the edges, cut holes. If you've used SketchUp you'll feel at home — and if you haven't, you'll be making real parts in minutes.
2. **Save it and print it.** One click checks it's really ready to print (no gaps, nothing floating in mid-air, small enough for your printer) and saves a file your slicer opens straight away.
3. **Starting an electronics project?** Choose your parts from over 2,000 ready-made boards, sensors, screens and modules. You don't need to know the part number: search for what it does, like *temperature*, *gyroscope* or *screen*, and add it to your box with one click.
4. **One click, one custom box.** The Enclosure Wizard arranges your parts and builds a printable case to hold them: holes lined up with the sockets, buttons and sensors, posts that print without supports, and a lid that fits.
5. **Not sure how to wire it up?** Click **Wiring**. A diagram and a tick-list show exactly which pin goes to which, with safe pins chosen for you and a warning wherever something needs care.
6. **We'll even write the code.** EvenUCan writes the program for your board. Install the free **ESPHome Device Builder** on your computer, click **Open in ESPHome**, and your device appears there: plug the board in and click Install, and ESPHome builds the program and puts it on the board.

> **Projects made simple:** your idea, a little thought, and some help from AI.

This manual follows the same path: drawing and printing first (chapters 6–8), then parts, the box, wiring and the code (chapters 9–11).

Rather be guided? The **step-by-step wizard** (chapter 10) walks you through all four steps, one question at a time, and checks the free software you need is installed.

2 Getting started

Which file to use

Everything is on the EvenUCan download page on GitHub: **github.com/bullbrand/EvenUCan/releases**. Pick the one for your computer. Each one is the whole program.

File	What it does
EvenUCan-Setup-1.9.42.exe	Windows: installs the program with a Start menu and desktop shortcut. Uninstall it from Settings › Apps.
EvenUCan-Portable-1.9.42.exe	Windows: just double-click it. Nothing is installed, so it also runs from a USB stick.
EvenUCan-1.9.42.dmg	Mac: open it and drag EvenUCan into Applications.
EvenUCan-1.9.42.AppImage **EvenUCan-1.9.42.deb**	Linux: the AppImage runs on its own (make it executable first); the .deb installs on Ubuntu, Debian, Mint and the like.
EvenUCan_1.9.42_Web.zip	The web-page version: unzip it and double-click **Start EvenUCan.bat** (Windows) or **start-here.sh** (Mac/Linux). This starts a small local server and opens the app in your browser. Keep all the files in the same folder.

Windows needs Windows 10 or 11 (64-bit). Installing a newer version over an older one keeps your drawings, parts and settings.

“Windows protected your PC” (and the Mac warning)

The first time you run it, Windows may show a blue box, because the program isn’t signed by a big company. Click **More info**, then **Run anyway**. On a Mac, right-click EvenUCan, choose **Open**, then **Open** again. You only need to do this once.

First thing to do: tell it your printer

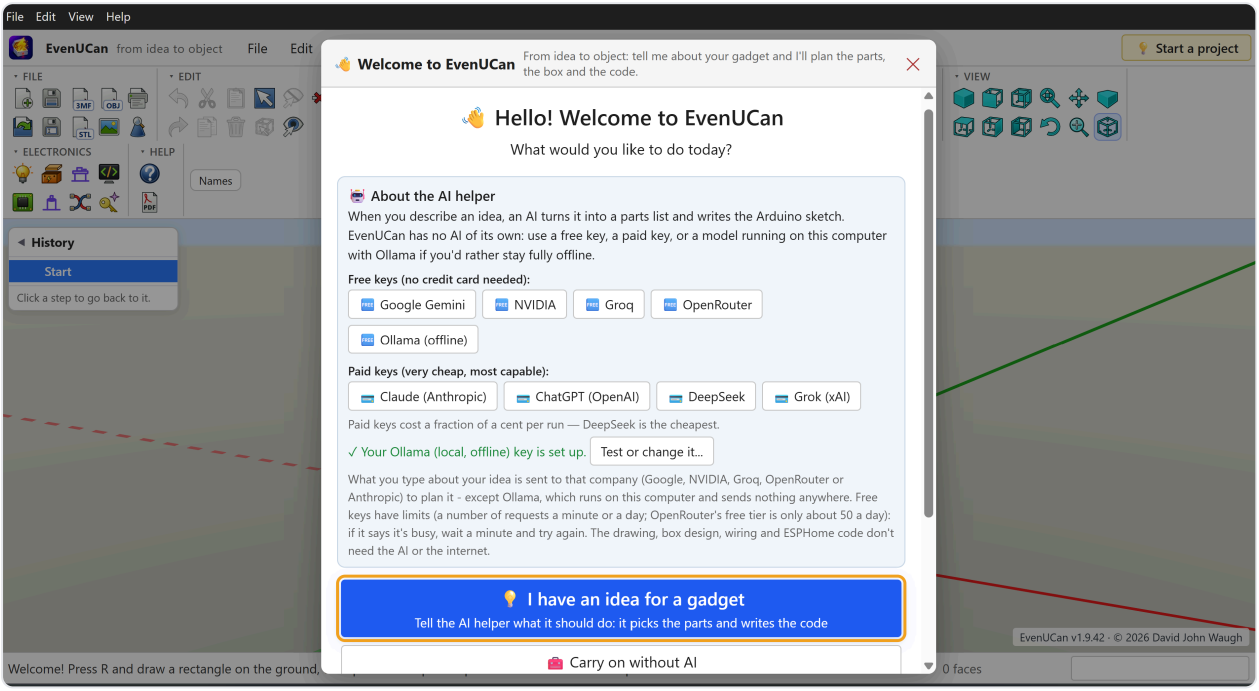
Click `Printer` on the top bar and pick your printer, or type its build size (X, Y and height Z, in mm). The grid then shows your bed, and the app warns you if anything is too big to print.

The step-by-step wizard NEW

EvenUCan walks you through a whole project in the right order, so you never end up writing code for parts you haven’t got, or wiring parts that don’t exist yet:

💡 Idea → 📦 Parts → 📦 Box → ⚡ Wiring → 💻 Code → ✅ **Done**

When you open EvenUCan with a fresh, empty drawing, the Welcome window greets you. Choose 💡 **I have an idea for a gadget** to be guided through the whole project, or close the window to go straight to the drawing canvas.



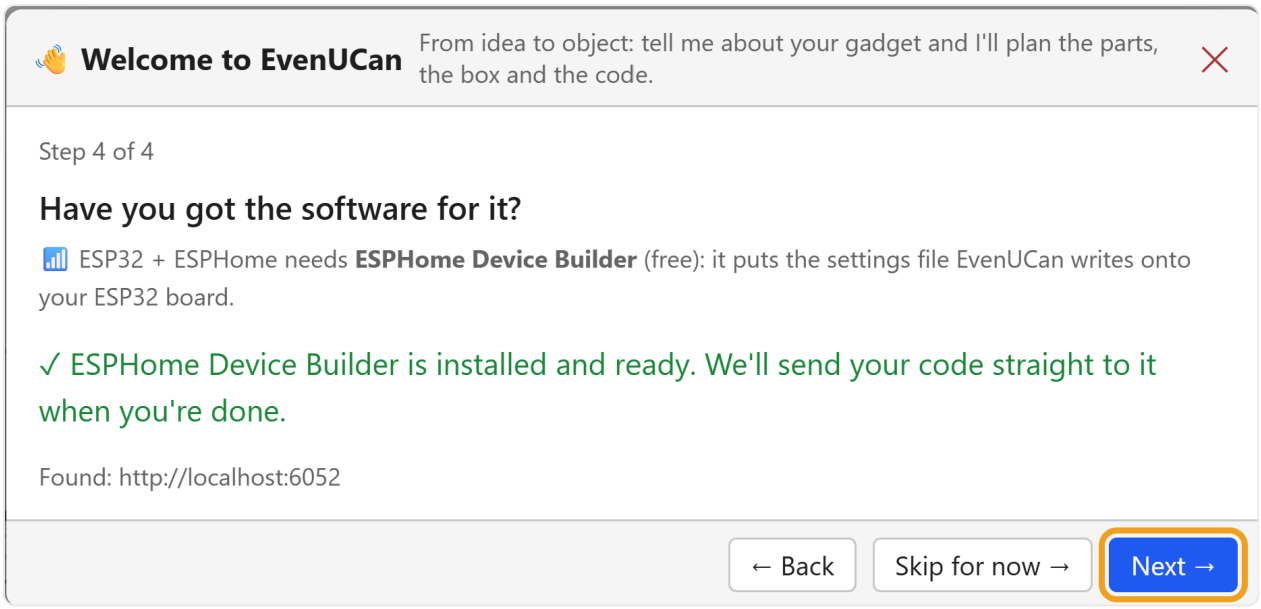
The Welcome window. The AI helper section lets you add your key; it collapses once a key is set.

The AI helper (and when it uses the internet)

The drawing, box design, wiring and ESPHome code all work offline. Only the AI helper needs the internet: the wizard's **I have an idea for a gadget**, **Write the code with AI**, and **Fill in automatically** in the Parts Library. It needs a key: a free one from Google Gemini, a free one from Ollama (local), your own paid Claude key, or a free or paid DeepSeek key.

Checking ESPHome is ready

The wizard's last setup step (Step 4 of 4) checks that ESPHome Device Builder is installed on your computer. If it finds it, you'll see a green tick and the address it's running on.



Step 4 of 4: ESPHome Device Builder is found and ready. Click **Next** to begin your project.

Carrying on without AI

Click **Carry on without AI**. You can start from an example project, or tick the parts you already have from the list. EvenUCan then designs the box, draws the wiring and writes the ESPHome code just as before, with no internet needed.

★ What’s new in 1.9.42

This chapter covers the newest parts of EvenUCan: DeepSeek AI support, automatic update notifications, and smarter AI parts matching.

DeepSeek AI key

EvenUCan now accepts a DeepSeek key as well as Google Gemini, Ollama and Claude. DeepSeek offers free API access (with rate limits) and paid plans. To add your key:

1. Go to **Tools › AI helper key** in the menu.
2. Choose **DeepSeek** from the provider list.
3. Paste your key (get one free at platform.deepseek.com) and click **Save**.

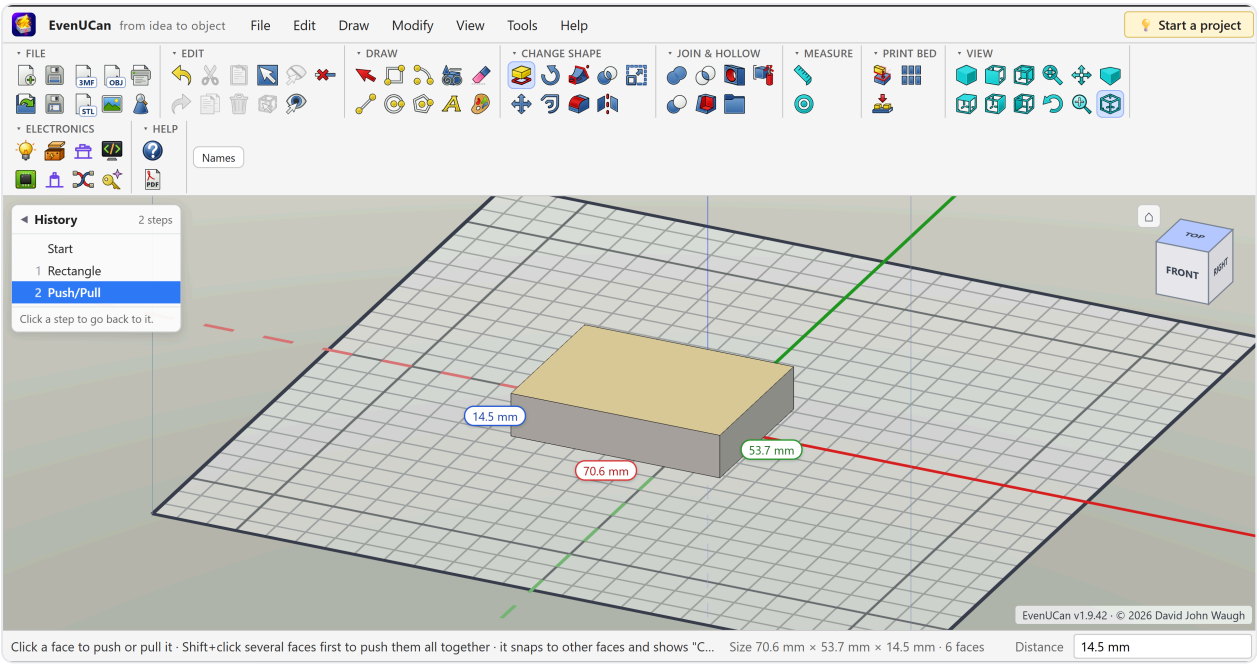
Update notifications

EvenUCan now checks quietly in the background whether a newer version is available. When one is, a small banner appears at the top of the window. Click it to go straight to the download page. You can carry on working — it never interrupts you.

Smarter AI parts matching

When you describe your gadget, the AI is now better at matching your description to the right part in the library. For example, describing a “smart LED strip” now reliably picks a NeoPixel-compatible module rather than a plain LED. The AI also double-checks its own suggestions against the parts list, so you spend less time correcting mismatches.

4 The screen



The main window showing a finished simple box. Menus along the top, the toolbar underneath, history panel on the left, and the size in the status bar at the bottom right.

Area	What it's for
Top bar	Files (New, Open, Save, Export), Undo/Redo, the view buttons (Iso, Top, Front...), Units, Resize, Printer, Check Model, Hide, Group, and the Enclosure, Parts, Wiring and Help windows. **About** shows which version you have.
Toolbar	The drawing tools, 3D Text, Shell, Hole, Flip and Lay flat, then the view tools, Group/Ungroup, Join, Subtract, Intersect, Hide/Unhide, Check Model, Enclosure and Parts. Hover over any button to see its name.
History panel	Shows every step in order. Click any step to jump back to it.
3D view	Your model on the printer bed. Red, green and blue lines are the X, Y and Z axes. The small **view cube** in the top right corner turns with the view.
Status bar	Left: a hint for the tool you're using. Right: the model's size.

The toolbar colours

The icon colours tell you what kind of tool it is: **blue** draws (Line, Rectangle, Circle, Arc); **orange** changes shapes (Push/Pull, Move, Rotate, Round, Offset, Follow Me); **red** erases

and **purple** paints; **teal** measures; **indigo** moves the view; **green** groups and checks; **amber** hides; **violet** opens the Enclosure Wizard and Parts Library.

Press `F1` or click `Help` at any time for a quick guide to every tool.

5 Moving around

Mouse action	What it does
Left button	Always belongs to the tool you've picked: it draws, pushes, moves... Click-click or press-drag-release both work.
Right button + drag	Orbit (turn the view) around whatever is under the mouse.
Right click	Opens the right-click menu (see below).
Middle button + drag	Pan (slide the view). Press the scroll wheel down and drag.
Scroll wheel	Zoom in and out, towards where the mouse points.

- `Shift` + `Z` (or the zoom-extents icon) fits everything in view.
- **Right-click** (without dragging) on the model opens a menu of common jobs: select the whole object, all or none, Delete, Hide, Group, Join, Subtract, Intersect, Flip / Mirror, Lay flat on this face, Round every edge, Zoom to selection, Check Model.
- The `Iso` `Top` `Front` `Right` `Back` `Left` buttons jump to standard views.
- **View** switches between *Perspective* (natural, far things smaller) and *Parallel* (no perspective, good for checking sizes by eye).

The view cube: choosing the side you draw on

The cube in the top right corner always turns with your view, so you can see which way you're looking.

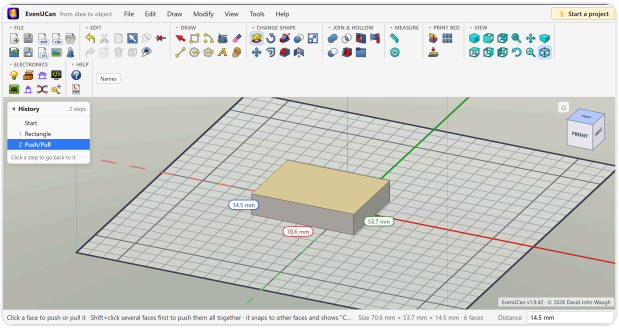
- **Click a face** of the cube (FRONT, RIGHT, TOP...) to look straight at that side. New shapes are then drawn on that side: a light blue grid shows it, and **Drawing on: Front** appears under the cube.
- **Click an edge or a corner** of the cube for an angled view. The side you draw on stays the same.
- **Drag the cube** to turn the view round, and click Δ (top left of the cube) to go back to the normal view of everything.

> **Tip:** When you start a new, empty drawing, the cube waits in the middle of the screen and asks which side you want to draw on. Click a side, or **Just start on the ground**, and it moves out of the way into its corner.

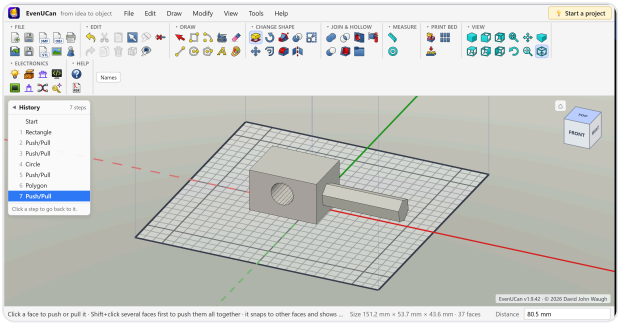
6 Drawing

EvenUCan works like SketchUp: draw a flat shape, then pull it up into 3D.

1. Press `R` and click two corners on the ground to draw a rectangle. To make it an exact size, type `40,30` and press `Enter`.
2. Press `P` (Push/Pull), click the rectangle and move the mouse up. Type `15` and press `Enter` for exactly 15 mm.
3. Draw a circle on the top face (`C`) and push it down through the box to cut a hole: when it reaches the far side it shows **Cuts Through**, so click to cut.



A Rectangle + Push/Pull makes a simple box in two steps.



Seven steps: Rectangle, Push/Pull, Circle, Polygon and more — building a detailed enclosure.

Tool	Key	How to use it
Select	Space	Click to select; Shift-click to add. Double-click selects a whole object. Drag left→right to select what's fully inside the box; right→left to select anything the box touches.
Line	L	Click points; closing the loop makes a face. A line across a face splits it.
Rectangle	R	Two corners. Type `width,depth` for exact sizes.
Circle	C	Centre, then radius. Type a radius, or `36s` to change the number of sides.
Push/Pull	P	Pull a face into 3D or push it in. Double-click repeats the last distance.
Move / Copy	M	Pick up and drop. Hold Ctrl to copy.
Rotate	Q	Centre, starting angle, end angle — or type a number and press Enter.
Round / Chamfer	J	Click a line where two faces meet to round it off. Type `3` for 3 mm rounded, `3c` for a 3 mm straight chamfer.
Offset	F	Click a face, then move in or out: a copy of its outline that far inside or outside it.

Pushing and pulling several faces at once

Hold Shift and click several faces first, then press P to push them all by the same amount at once — handy for making a tray with walls all the same height.

Undoing mistakes

Press Ctrl + Z to undo, as many times as you like. The **History** panel on the left lists every step, so you can jump back several steps at once — just click the step you want to go back to.

7 Making it ready to print

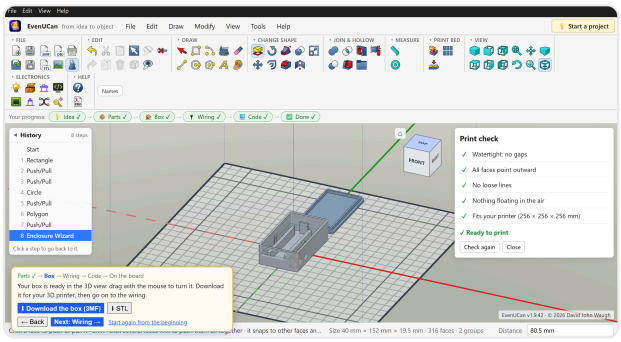
Before you send something to your printer it’s worth asking EvenUCan to check it over.

Check Model

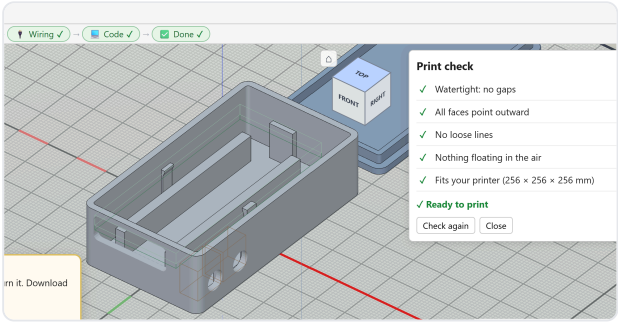
Click **Check Model** (the green tick on the toolbar, or the top bar). EvenUCan looks for the common problems that stop a print from working:

- **Holes** — gaps in the surface where it should be solid.
- **Reversed faces** — faces pointing the wrong way in (fix them with right-click › Reverse faces).
- **Too big for your printer** — it warns you if the model is bigger than the bed you set up.
- **Floating pieces** — parts that aren’t attached to anything.

If everything is fine, it says **Ready to print** in green. If not, it lists the problems and you can click on them to zoom in.



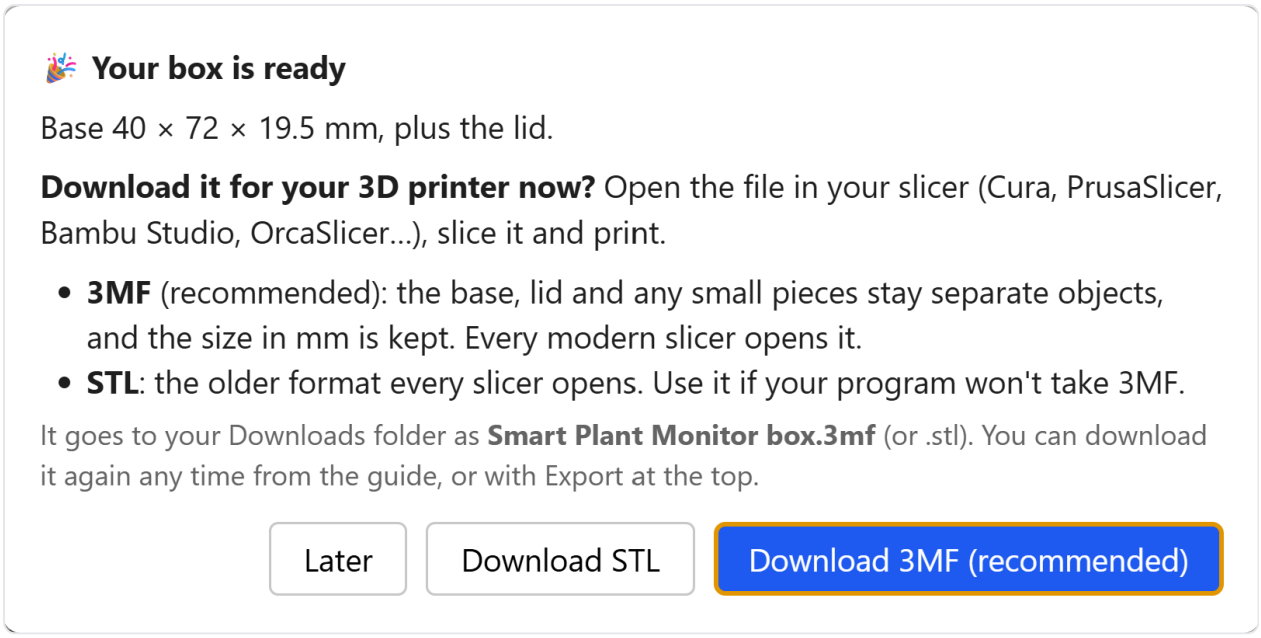
All five checks pass: **Ready to print**. The box is watertight and fits the printer bed.



Close-up of the finished enclosure. Internal rails, posts and cutouts are all built by the wizard.

Exporting for your slicer

Use **Export 3MF** (recommended) or **Export STL** at the top. 3MF keeps the size in mm and keeps separate objects separate, so your slicer can arrange them. STL works with every slicer. Open the file in Cura, PrusaSlicer, Bambu Studio, OrcaSlicer or any other slicer, slice it and print.



The download dialog after the Enclosure Wizard builds your box. 3MF is recommended; STL works if your slicer won't take 3MF.

> **Tip:** If your slicer won't take 3MF, use STL instead — they both work the same way inside the slicer.

Lay flat

Right-click a face and choose **Lay flat on this face** to tip your model so that face sits on the ground. This is useful for making the best flat side face down before you export.

8 Saving and opening

EvenUCan saves your work as a **.3mf** file. This is also the file your slicer opens — so you only ever deal with one file per project. Use **Save** (Ctrl + S) often.

The file keeps your drawing, your electronics settings, your parts list, the wiring, and the ESPHome code all in one place. Open it somewhere else (another PC, a reinstall) and all of that comes back too.

> **Autosave:** EvenUCan saves a backup every few minutes. If it closes unexpectedly, it offers to restore your work next time you open it.

New drawing

Click **New** at the top bar (or Ctrl + N). If you have unsaved changes, it asks first.

Opening a file

Click **Open** (or Ctrl + O) and pick your .3mf file — or just drag a .3mf or .stl file from your file manager into the window.

Exporting an image

Use **Export PNG** at the top bar to save a picture of whatever is currently on screen — handy for sharing a preview of your design.

9 The Parts Library

Over 2,000 boards, sensors, screens, and modules — ready to drop into your project. You don't need to know part numbers. Search for what it does.

Finding a part

1. Click **Parts** in the top bar (or **Tools › Parts library**).
2. Type what the part does in the search box — for example *temperature*, *oled display*, or *motor driver*.
3. Click a result to see a picture and its key measurements.
4. Click **Add to project** to place it on the canvas. EvenUCan places it flat on the ground.

> **Tip:** The shape EvenUCan places is the actual footprint of the board — the right size, with the connector positions marked. The Enclosure Wizard uses these positions to cut the right holes in your box.

Filling in a part automatically

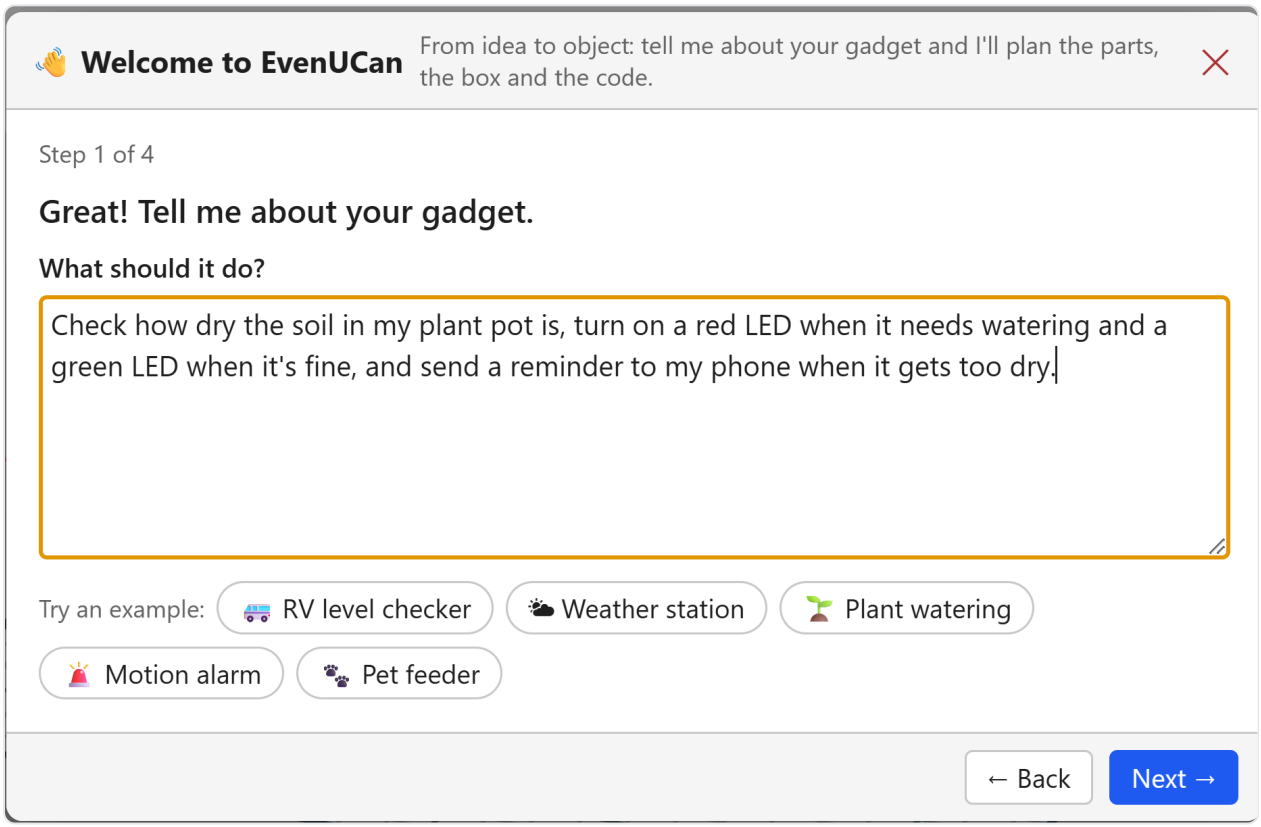
If you have an AI key set up, click **Fill in automatically** in the Parts Library and describe the part in your own words. EvenUCan searches the library for the closest match and adds it for you — handy when you know what a part does but not its exact name.

10 The Enclosure Wizard

Tell EvenUCan what you want to build, and it picks the parts, designs the box and writes the code — all in one guided flow.

Step 1: describe your gadget

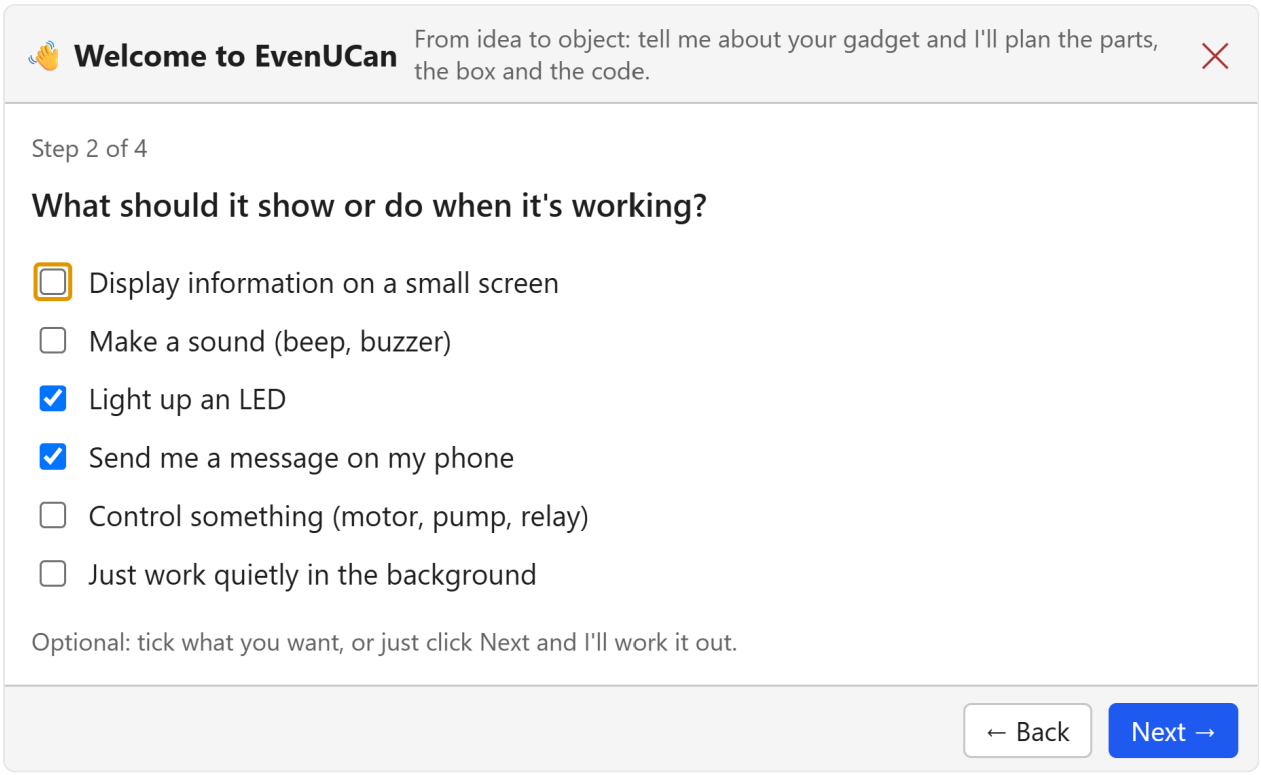
Click 💡 **Start a project** (top right) to open the wizard. At Step 1 of 4, type what you want your gadget to do in plain language. Use the example buttons below the box if you'd like to start from a template.



Step 1 of 4: describe your gadget in your own words. Example projects are shown below if you need inspiration.

Step 2: choose its features

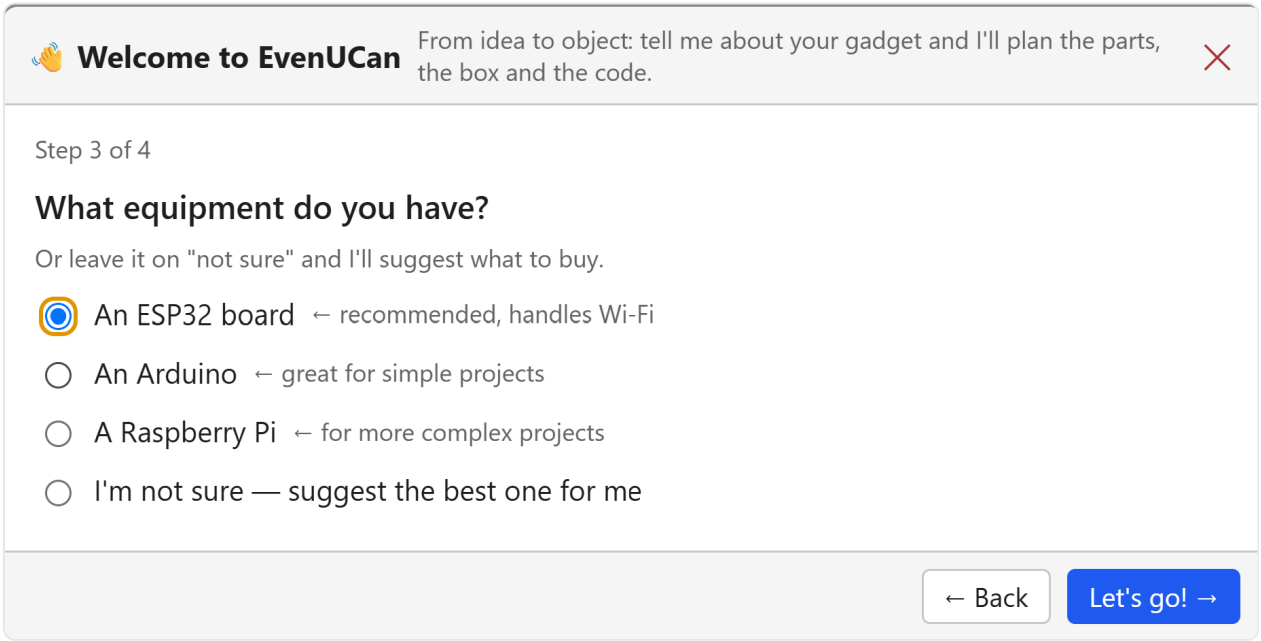
Tick what your gadget needs to do when it's working — indicators, messages to your phone, and so on. EvenUCan uses these answers to pick the right parts.



Step 2 of 4: tick what the gadget should do. EvenUCan uses these answers when choosing parts.

Step 3: choose your hardware

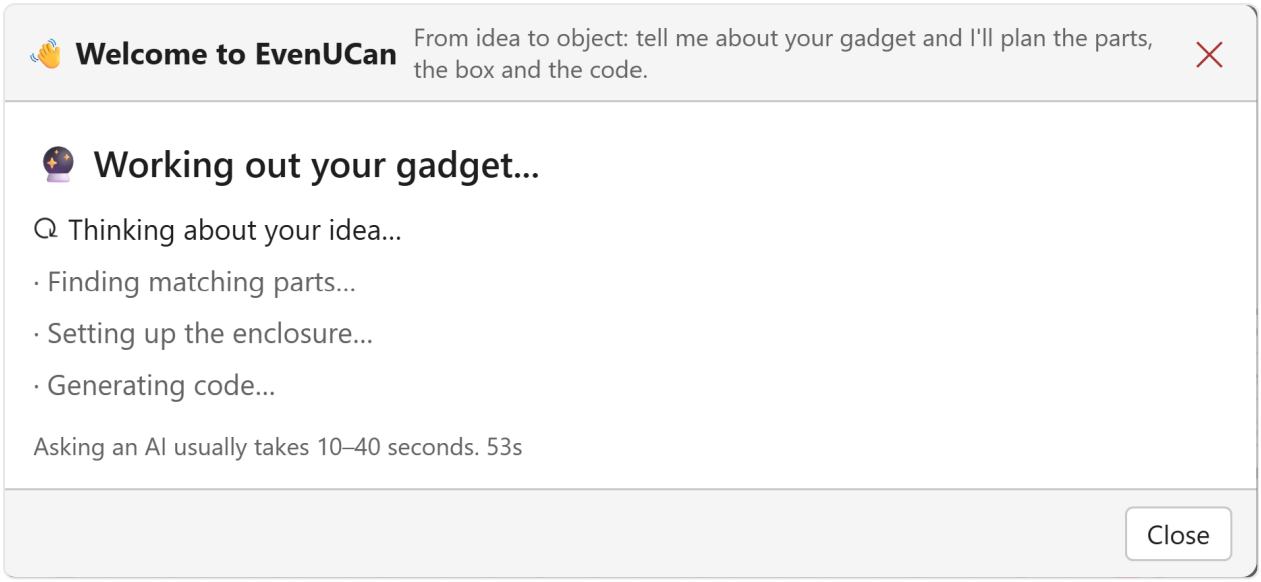
Tell the wizard what board you have — or leave it on **I'm not sure** and EvenUCan will suggest the best one for your project. ESP32 + ESPHome is recommended for anything that needs Wi-Fi.



Step 3 of 4: pick your board. ESP32 is recommended for Wi-Fi projects. Choose "I'm not sure" to get a suggestion.

The AI works out your project

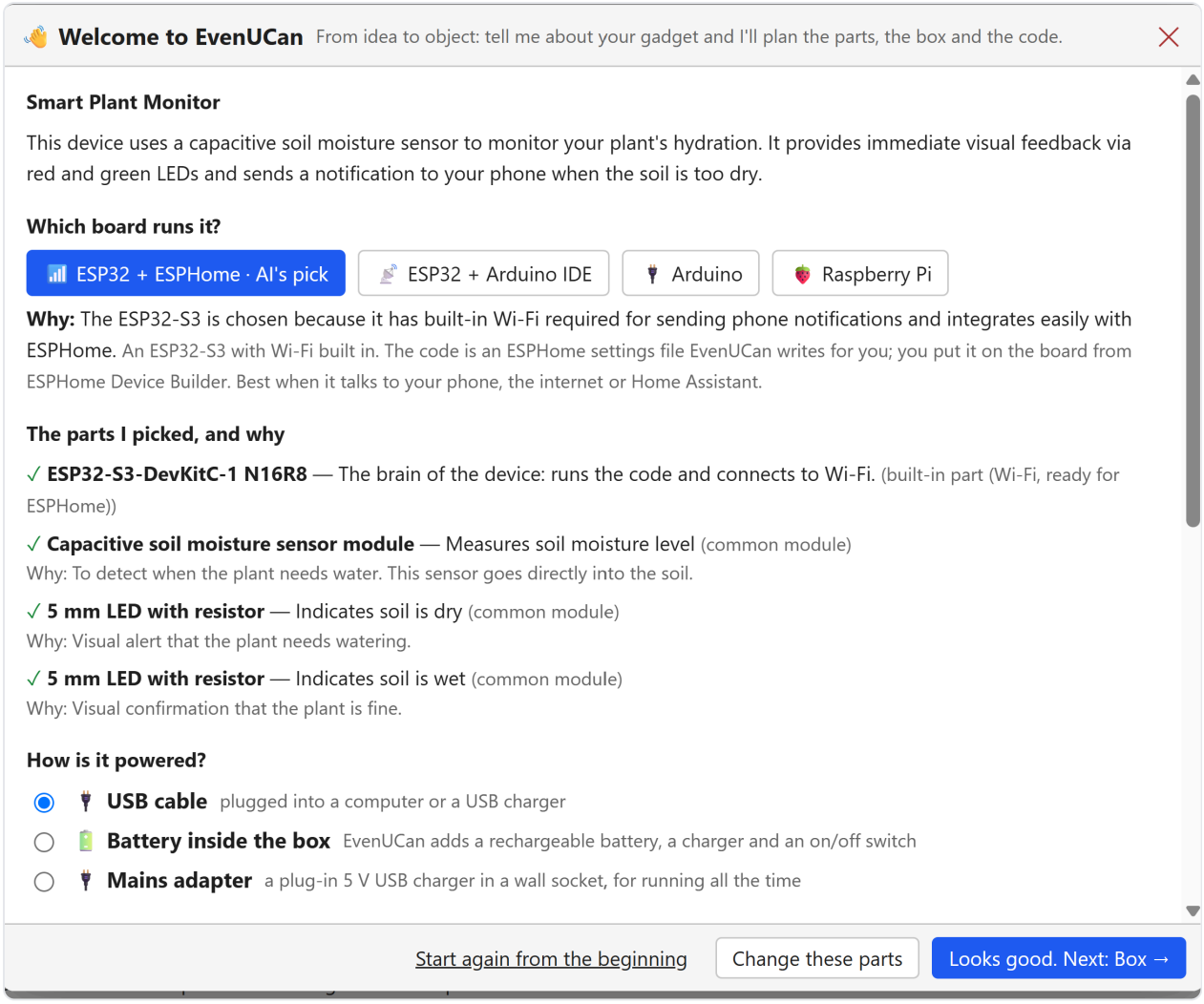
After Step 4, EvenUCan asks your AI to plan the whole project. This takes 10–40 seconds while it thinks about the parts, the box and the code.



The AI thinks through your idea, finds matching parts, sets up the enclosure and generates the code. Usually 10–40 seconds.

Reviewing the AI's suggestions

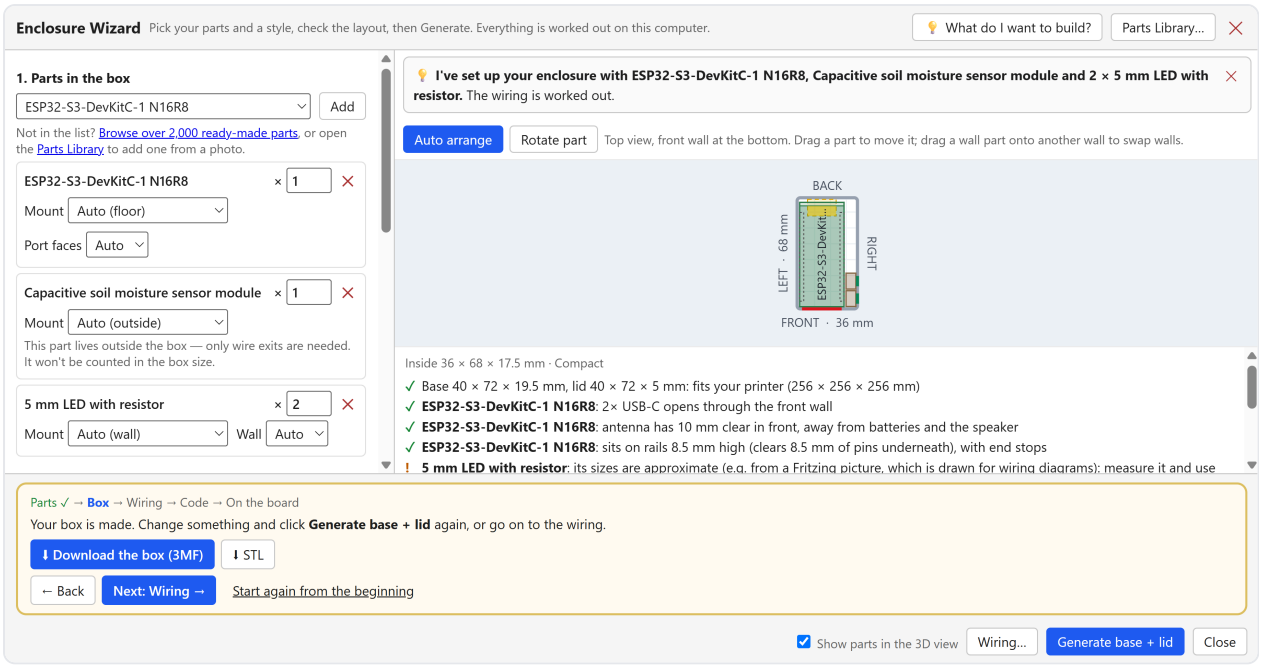
When the AI finishes, it shows you its plan: the board it chose, every part and why, and how the gadget will be powered. You can switch to a different board, change the power option, or click **Change these parts** to swap anything out.



The AI explains its choices. You can accept the plan or change any part before moving on.

The Enclosure Wizard layout

Once you accept the parts, the Enclosure Wizard arranges them in the box. The top-view diagram shows where each part will sit. Drag a part to move it; use **Auto arrange** to let EvenUCan find the most compact layout. When the layout looks right, click **Generate base + lid**.



The Enclosure Wizard: parts list on the left, top-view layout on the right. Click **Generate base + lid** when you're happy with the arrangement.

The step-by-step guide bar

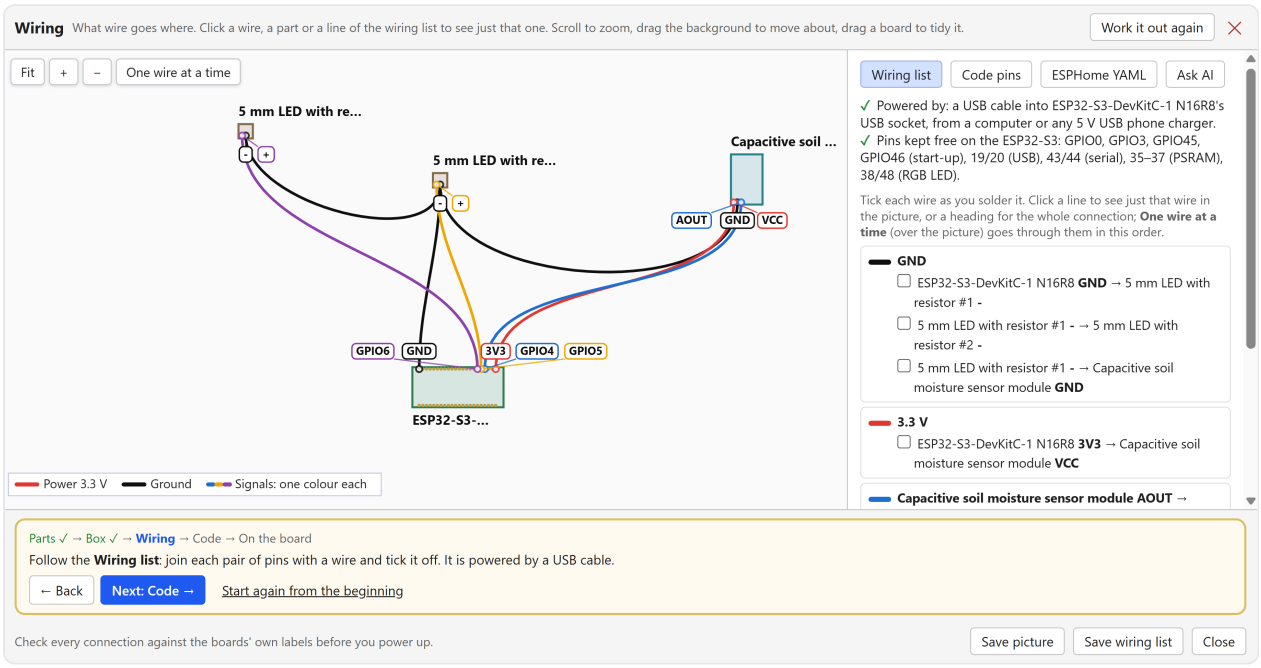
After the box is generated, a yellow guide bar appears at the bottom. It tells you what to do next — download the box, check the wiring, put the code on the board — and ticks each step off as you go. Click any step to jump straight to it.

> **Tip:** If a hole is in the wrong place, drag the part to the right position inside the box and click **Regenerate** — the box rebuilds around the new layout.

11 Wiring

Click **Wiring** in the top bar to open the wiring diagram. EvenUCan shows you which pin on your board connects to which pin on each part, picks safe pins for you, and warns you wherever something needs care.

Reading the diagram

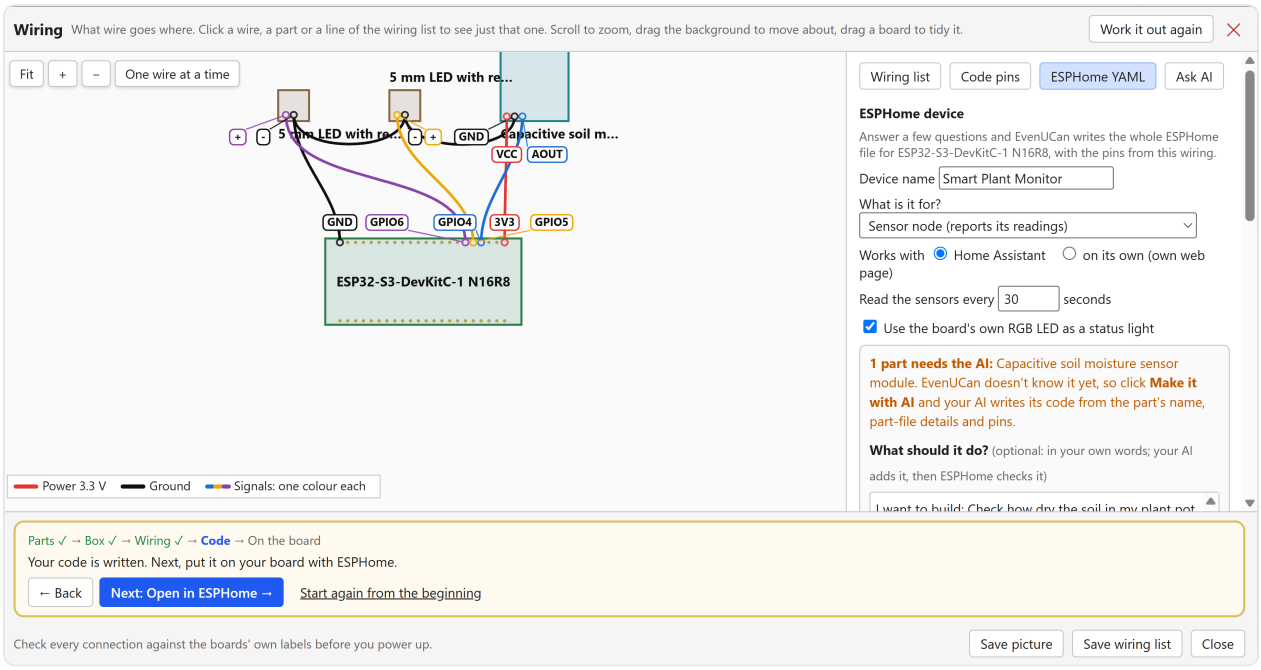


The wiring diagram: each part is shown with its connections. The wiring list on the right lets you tick off each wire as you solder it.

- **Wire colours:** red for power (3.3 V), black for ground, and a different colour for each signal. A small key explains them.
- **Focus:** click a row in the wiring list, a wire, or a part. Everything else fades and one big label says exactly what to connect.
- **One wire at a time:** click `One wire at a time`, then `Next` and `Back` to work through the connections in order. Tick each wire off as you solder it.
- **Zoom and move:** use the mouse wheel to zoom, drag the background to move around, and click `Fit` to see the whole diagram.
- **Save picture** saves a sharp picture of the whole diagram, whatever you're zoomed into.

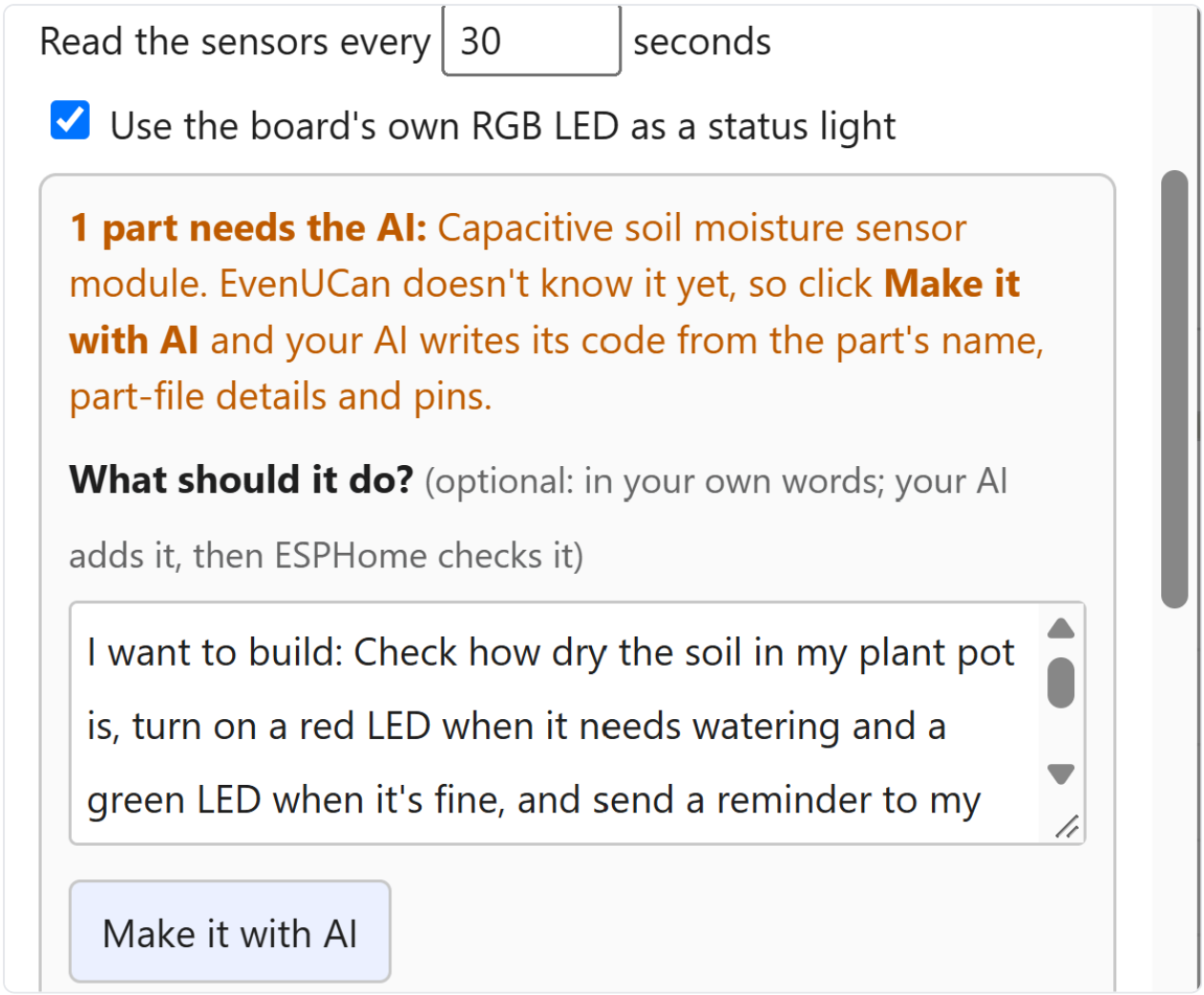
The ESPHome YAML tab and AI code generation

Switch to the **ESPHome YAML** tab in the wiring window to configure the code. Set a device name, choose whether it works with Home Assistant or on its own, and set how often the sensors read.



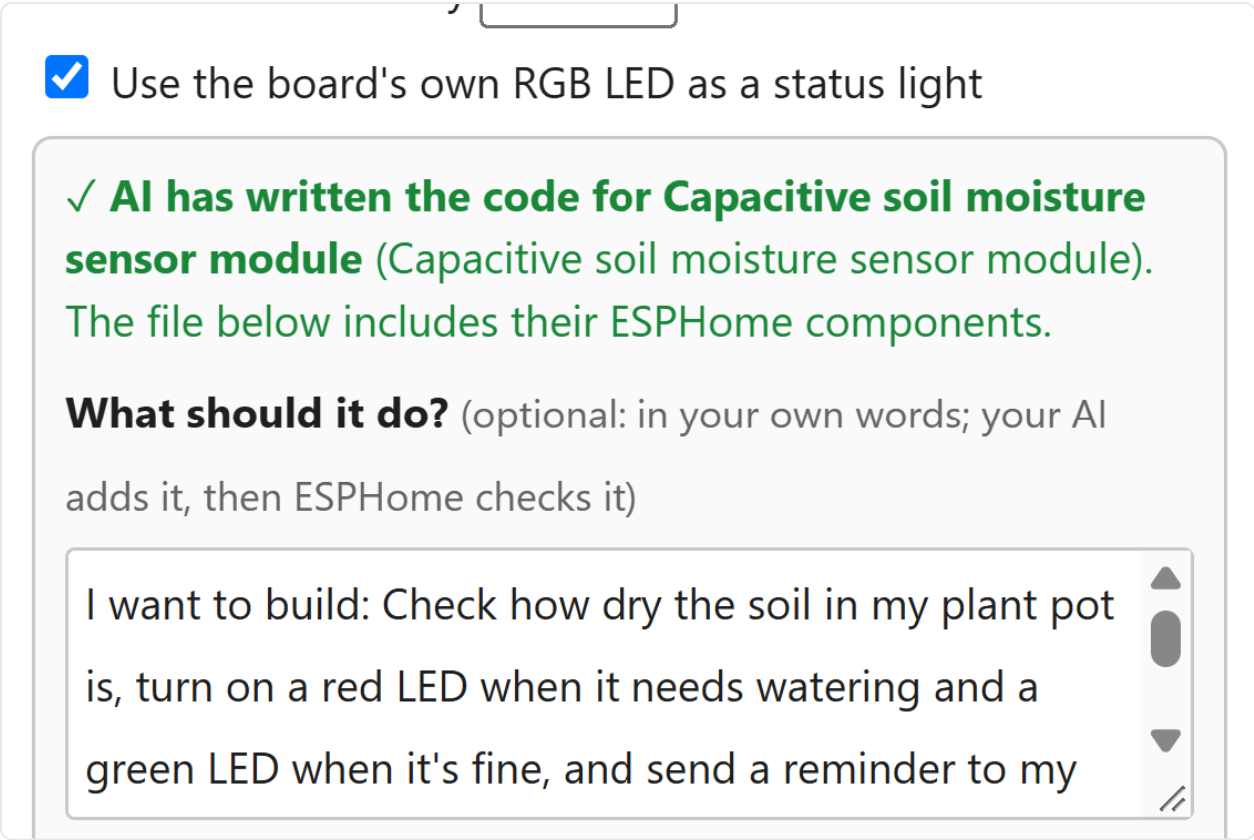
The **ESPHome YAML** tab: set the device name and options, then click **Make it with AI** for any part that needs custom code.

If a part isn't in EvenUCan's code library yet, an orange notice tells you: **1 part needs the AI**. Click **Make it with AI** and your AI writes the ESPHome code from the part's name, details and pins.



When a part needs custom code, an orange notice appears. Click **Make it with AI** and the code is written for you.

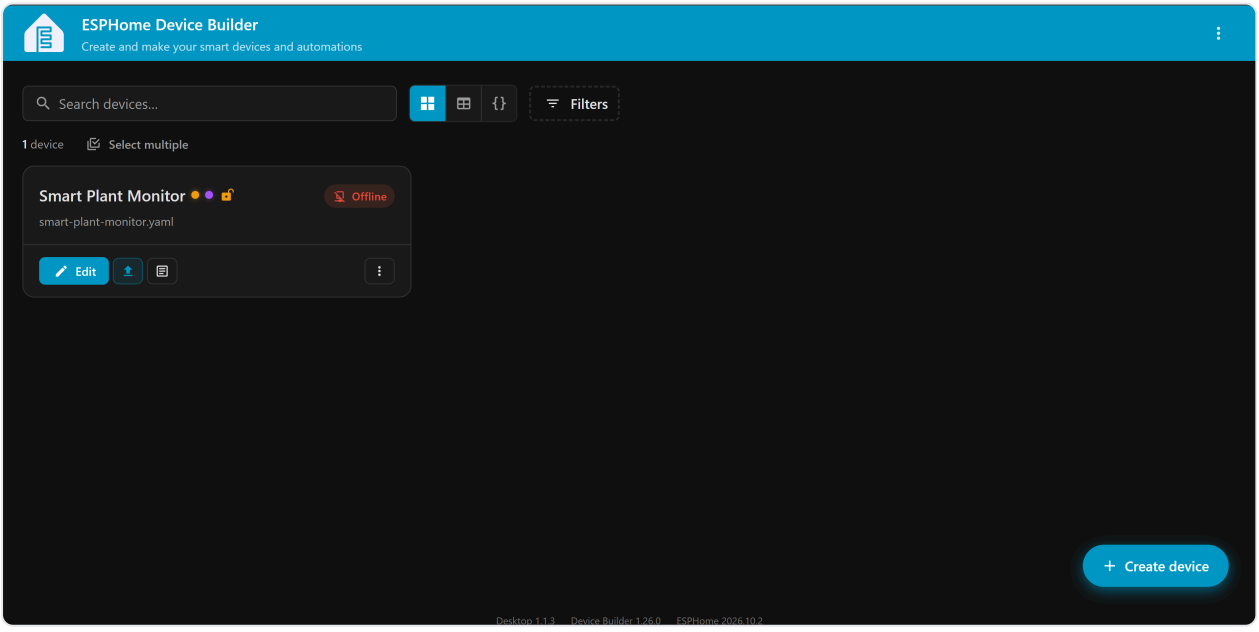
When the AI finishes writing the code, a green tick confirms it’s done and the ESPHome file now includes that part’s components.



Green tick: the AI has written the code for the soil moisture sensor. The ESPHome file is updated and ready.

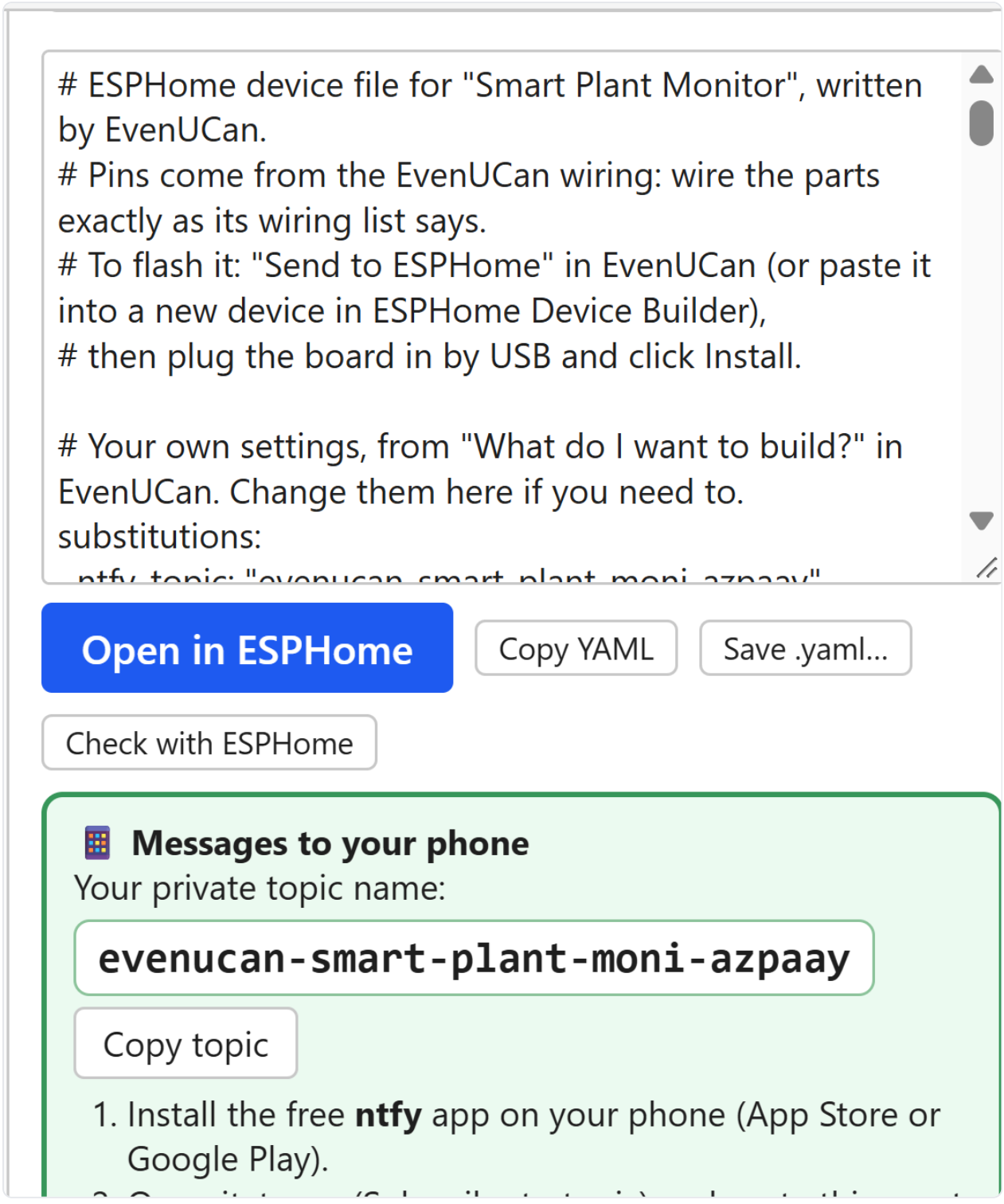
Sending the code to your board

Click **Open in ESPHome**. The ESPHome Device Builder opens with your device already listed. Plug the board in by USB, then click **Install** inside ESPHome. It compiles the code and loads it onto the board.



ESPHome Device Builder with the new device. Click **Edit** to review the code or the upload button to flash it to the board.

You can also copy the YAML or save it as a .yaml file and paste it into a new device in ESPHome yourself.



The finished ESPHome file with phone-notification settings. Copy the private topic name and subscribe to it in the ntfy app on your phone.

> **Wi-Fi:** your Wi-Fi name and password are never put in the code by EvenUCan. For ESPHome they come from ESPHome's **Secrets**; for Arduino the guide shows where to type them.

Messages to your phone

If your gadget sends alerts, it uses the free **ntfy** service. No account is needed:

1. Install the free **ntfy** app on your phone (App Store or Google Play).
2. Open it, tap **+** (Subscribe to topic) and paste your private topic name, shown in the YAML tab and the guide. Leave the server as ntfy.sh.
3. That's it: when your gadget sends a message, it pops up on your phone.

Anyone who knows the topic name can read the messages, so keep it to yourself.

12 Sharing with other people

Your .3mf file has everything in it: the drawing, the parts, the wiring, the code. Send the file and they can open it in EvenUCan and pick up exactly where you left off.

To share just the 3D model (without the electronics settings), use **Export STL** or **Export OBJ**. These work in any 3D program, not just EvenUCan.

To share a picture of your design, use **Export PNG** for a screenshot of the current view, or **Save picture** in the Wiring window for the wiring diagram.

13 Keyboard shortcuts

Hover over any toolbar button to see its shortcut key shown in the tooltip. Here are the most useful ones:

Key	What it does
Space	Select tool
R	Rectangle
C	Circle
L	Line
A	Arc
P	Push/Pull
M	Move / Copy
Q	Rotate
J	Round / Chamfer
F	Offset
Ctrl + Z	Undo
Ctrl + Y	Redo
Ctrl + S	Save
Ctrl + N	New drawing
Ctrl + O	Open file
Shift + Z	Zoom to fit everything
Delete	Delete selected
G	Group selected
U	Join into one solid
S	Subtract (cut one from another)
F1	Quick guide (all tools and shortcuts)

You can also type a size any time a tool is waiting for a measurement — just type the number and press Enter. For two dimensions, type them separated by a comma: 40,30 for 40 mm by 30 mm.

14 Problems and answers

Problem	What to do
Windows says “Windows protected your PC”	Click **More info**, then **Run anyway**. You only need to do this once.
Mac says the app is from an unidentified developer	Right-click EvenUCan, choose **Open**, then **Open** again. Once only.
The model looks wrong in my slicer	Run **Check Model** first. If it finds reversed faces, right-click them and choose **Reverse faces**. Try 3MF instead of STL.
The AI won’t work — it asks for a key	Go to **Tools › AI helper key** and add a free Gemini or DeepSeek key. You can always click **Carry on without AI** to skip it.
Some AI providers say “no internet” in the desktop app	Providers like DeepSeek and Groq go through a direct connection. Update to v1.9.42 or later, which adds direct-fetch support for all providers.
I closed the wizard	Click 💡 **Start a project** at the top right, or any step in the **Your progress** strip.
The box holes are in the wrong place	Drag the part to the right position inside the box and click **Regenerate**.
It says “Add some parts first”	Wiring and code need parts. Pick them in the Enclosure Wizard or the Parts Library first.
No messages on my phone (ntfy)	Check the topic in the ntfy app matches exactly (use **Copy topic**), and that the gadget is on your Wi-Fi.
The web version fails to connect to AI	Open EvenUCan using **Start EvenUCan.bat** (Windows) or **start-here.sh** (Mac/Linux) instead of opening the HTML file directly. This gives it a proper web address so AI calls can reach the internet.
I can’t find a part in the library	Try different search words. If you have an AI key, use **Fill in automatically** and describe the part in your own words.

> **Still stuck?** Click **Help › EvenUCan on GitHub** to see the issue tracker — someone may have had the same problem. Or open a new issue and describe what happened.

15 Credits

EvenUCan is written and maintained by **David John Waugh**.

It would not exist without these fine open-source projects:

- **three.js** — 3D rendering in the browser (threejs.org)
- **ESPHome** — the firmware builder that turns the code into a running device (esphome.io)
- **Google Gemini API** — free AI for gadget ideas and code
- **DeepSeek API** — additional free AI option
- **NVIDIA NIM API** — another free AI option for the desktop app
- **Anthropic Claude API** — powerful AI for detailed projects
- **ntfy** — the free push-notification service (ntfy.sh)

Parts data comes from the manufacturers' own datasheets and from the community. Thank you to everyone who has contributed a part, filed a bug report, or suggested an improvement.

EvenUCan is free software, released under the **GNU General Public Licence v3** or any later version. The full licence text is at gnu.org/licenses/gpl-3.0.html.

> **Source code:** github.com/bullbrand/EvenUCan